

Appendix BCT-SMOKE

OHC Resonance Frequencies and Void-Geometry Filter Specifications for Selective Pyrolysis Carcinogen Suppression

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App BCT-SMOKE.1 — BCT Geometric Inputs

Locked constants (v47k):

$$r_{\text{oct}} = (\sqrt{2} - 1)/2 = 0.20710678\dots$$

$$r_{\text{tet}} = (\sqrt{6} - 2)/4 = 0.11237244\dots$$

$$\alpha_0 = r_{\text{oct}} \cdot r_{\text{tet}} / \pi = 0.00740813\dots$$

$$\omega_{\text{J}^2} = 1/\pi \Rightarrow \omega_{\text{J}} = 0.56419\dots \text{ (BCT units)}$$

$$r_{\text{oct}} / r_{\text{tet}} = 1.84306\dots \text{ (selectivity ratio)}$$

$$1 + r_{\text{oct}} = 1.20711\dots \text{ (pore geometry factor, App AB2)}$$

App BCT-SMOKE.2 — PAH Ring-Closure Coupling Frequency

Benzo[a]pyrene ring-closure transition state geometry: nascent hexagonal C₆ arrangement, sp² hybridisation, C–C forming bond $\approx 1.40 \text{ \AA}$. In BCT units scaled to Bohr radius $a_0 = 0.529 \text{ \AA}$:

$$d_{\text{aromatic}} / a_0 = 1.40 / 0.529 = 2.646 a_0$$

The hexagonal ring spans $2\pi/r_{\text{oct}}$ void-radius units:

$$N_{\text{void}} = 2\pi / r_{\text{oct}} = 2\pi / 0.20711 = 30.34$$

The OHC coupling harmonic number for PAH ring closure:

$$n_{\text{PAH}} = \text{round}(N_{\text{void}} / (2\pi)) = 5 \text{ (fifth harmonic)}$$

PAH coupling frequency (fifth harmonic):

$$\omega_{\text{PAH}} = \omega_{\text{J}} \times n_{\text{PAH}} / r_{\text{oct}} = 0.56419 \times 5 / 0.20711$$

$$\omega_{\text{PAH}} = 13.63 \text{ (BCT units)}$$

Converting to SI ($\omega_{\text{P}} = 1.855 \times 10^{43} \text{ rad/s}$):

$$\omega_{\text{PAH}}(\text{SI}) = 13.63 \times 1.855 \times 10^{43} / (2\pi) \text{ Hz}$$

$$f_{\text{PAH}} \approx 40 \text{ THz } (\lambda \approx 7.5 \text{ }\mu\text{m, mid-IR})$$

This falls in the C=C aromatic out-of-plane bending region ($700\text{--}900 \text{ cm}^{-1}$ and $1400\text{--}1600 \text{ cm}^{-1}$).
CANDIDATE — harmonic order requires experimental determination. Range: $15\text{--}84 \text{ THz}$ depending on harmonic.

App BCT-SMOKE.3 — TSNA Coupling Frequency

N-nitroso bond (N–N=O) formation geometry: tetrahedral void scale. TSNA coupling frequency:

$$\omega_{\text{TSNA}} = \omega_{\text{J}} / r_{\text{tet}} = 0.56419 / 0.11237 = 5.02 \text{ (BCT units)}$$

$$f_{\text{TSNA}} \approx 14.8 \text{ THz } (\lambda \approx 20 \text{ }\mu\text{m}, \text{ far-IR})$$

Consistent with N–N stretch and N=O stretch frequencies in nitrosamines (1000-1500 cm⁻¹).
CANDIDATE.

App BCT-SMOKE.4 — Nicotine/Cannabinoid Preservation Frequency

Nicotine pyridine ring N–C coupling — must NOT be disrupted:

$$\omega_{\text{nicotine}} = \omega_{\text{J}} \times r_{\text{oct}} = 0.56419 \times 0.20711 = 0.1168 \text{ (BCT units)}$$

$$f_{\text{nicotine}} \approx 0.34 \text{ THz } (\lambda \approx 880 \text{ }\mu\text{m}, \text{ THz range})$$

This is well separated from f_PAH (40 THz) and f_TSNA (15 THz) by factors of 117× and 44× respectively. Selective intervention is geometrically justified — the frequencies are widely separated.

App BCT-SMOKE.5 — Filter Pore Geometry Derivation

Molecular kinetic radii (standard values):

Molecule	MW (g/mol)	Kinetic radius (Å)	BCT prediction	Transmission
Nicotine	162	3.50	$r_{\text{oct}} \cdot a_0 \cdot 33 \approx 3.62 \text{ Å}$	PASS
THC	314	3.80	within (1+r_oct) window	PASS
CBD	314	3.90	within (1+r_oct) window	PASS
Benzo[a]pyrene	252	5.10	> R_pore → blocked	BLOCKED
Chrysene	228	4.90	> R_pore → blocked	BLOCKED
NNK (TSNA)	208	4.60	> R_pore → blocked	BLOCKED

Table A1: Molecular transmission predictions for BCT-SMOKE Mechanism 2 filter. R_pore = 4.2 Å. CANDIDATE — experimental verification required.

App BCT-SMOKE.6 — Connection to App AB2

The pore geometry factor (1 + r_oct) = 1.20711 appearing in the filter specification R_pore = R_nicotine·(1+r_oct) is the same factor that appears in the 1/32 bilinear normalisation (App AB2):

$$1/32 = (1/|Q_8|) \cdot r_{\text{oct}}(1+r_{\text{oct}}) = (1/8) \cdot (1/4)$$

App AB2 was closed June 4, 2026 via the D4 8v Dynkin analysis (Letter 249). The (1+r_oct) factor is therefore geometrically grounded in the D4 root system, not empirically fitted. The BCT-SMOKE filter pore geometry inherits this derivation: the filter selectivity is a D4 root system prediction.

App BCT-SMOKE.7 — Tier Assessment

<i>Component</i>	<i>Tier</i>	<i>Basis</i>
BCT void geometry inputs	PROVEN	Locked v47k constants
Molecular bond angle framework	PROVEN	Letter 80, locked
(1+r_oct) pore factor	PROVEN	App AB2 closed June 4 2026
OHC coupling mechanism	CANDIDATE	Same as BCT-PFAS, BCT-HIVE
PAH ring-closure coupling	CANDIDATE	Harmonic order unverified
Specific candidate frequencies	CANDIDATE	Experimental required
Filter selectivity predictions	CANDIDATE	Pore geometry unverified
Combined >80% carcinogen reduction	CANDIDATE	Experimental required

Table A2: Honest tier assessment. PROVEN items are locked BCT constants. CANDIDATE items require experimental verification.